2023-2024 秋冬 数字逻辑设计 回忆卷

填空题

1. How many stable states does a flip flop have? How many flip flops are needed to represent an 8-bit binary number?

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2. (6B)$_H$ = ( )$_{10}$ = ( )$_2$

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3. Write the Sum of Minterms (SOM) for the following formula:

$F = A \oplus B \oplus 1$

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4. Given an 8-bit multiplier with inputs D0~D7 = 10110010. When the select signal changes from 000 to 111, what's the output sequence?

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5. A 7-segment display is connected using common cathode. To display 5, the input from a to g should be?

选择题

1. Given a $2^{22} - 8bits$ DRAM, which of the followings are false:

- A. There are 4M addresses.
- B. The capacity of the DRAM is 8M bytes.
- C. When using a 12-4096 decoder for row addresses, we should use a 1024-1 multiplexer for column addresses.
- D. Forgotten

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2. 2 flip-flops and a combinational circuit are given in the figure (not shown), What's the maximum frequency of the circuit.

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3. Which of the following codes is not weighted:

- A. 8421BCD
- B. 5421BCD
- C. Excess-3 code
- D. 2421BCD

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4. Given $F = \sum(0, 1, 2, 4)$, how to represent F in maxterms?

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5. A PAL is given in the figure (not shown), what's the output equation?

化简题

1. Given a F formula represented by A, B, simplify to sum of product form.

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2. Given $F = \sum(1, 4, 5, 6, 7) + d\sum(8, 9, 10, 11, 12, 13)$, simplify F. (Specific numbers forgotten)

编程题

Use Verilog behavioral description to form the circuit given in problem “选择.2”

设计题

1. Given 3 4-bit binary counters, design a day counting circuit from 001 to 365 and repeat the process, each of the counters in BCD code.

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2. Design a majority voter circuit with a 3-8 decoder, if more than 2 people vote, output 1.

- (1) Find truth table
- (2) Boolean formula
- (3) Draw the circuit

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3. Design a serial multiplier, X is the input and Z is the output, let every Z = 5X. 3 D flip-flops and a 1-bit full adder are given.